# Algorithmic Fairness Testing & Mitigation Report

*Prepared by: Data Science & AI Verification Team*
*Status: COMPLIANT (Post-Mitigation)*

## 1. Testing Methodology & Experimental Design

This report details the technical testing process and performance validation of the bias mitigation strategy applied to our automated screening tool. To establish a controlled baseline, we generated a synthetic dataset of 1,000 candidate profiles. The dataset simulates standard features (Years of Experience, Standardized Technical Test Score, and Education Level) alongside a protected demographic attribute (Gender: Male/Female).

To replicate historical human recruiter bias, we constructed the target label *'Hired'* using a latent qualification score where female applicants faced a penalizing coefficient. This means that a female candidate with the exact same experience, test score, and degree as a male candidate was assigned a significantly lower statistical probability of being marked as 'Hired' in the training set.

| Feature Name | Data Type | Description | Distribution / Modeling |
|---|---|---|---|
| Gender (Protected) | Categorical | Protected group marker | Binomial (50% Male / 50% Female) |
| Experience | Numerical | Years of relevant job experience | Uniform distribution between 1 and 15 years |
| Test Score | Numerical | Score on technical entrance test | Normal distribution (mean=70, sd=15), clip [30, 100] |
| Education Level | Categorical | Highest degree achieved | Bachelor (50%), Master (30%), PhD (20%) |
| Hired (Target) | Binary | Historical screening decision | Bernoulli trial based on qualifications + gender bias |

## 2. Fairness Definitions & Metric Equations

We evaluated three key mathematical definitions of fairness to assess discrimination:

1. **Disparate Impact (DI) Ratio:** Compares the selection rate of the unprivileged group to the privileged group. Crucially, the US Equal Employment Opportunity Commission (EEOC) enforces the *'four-fifths rule'*: a ratio below **0.80** indicates illegal discrimination.
   \[DI = rac{P(\hat{Y}=1 | ext{Female})}{P(\hat{Y}=1 | ext{Male})}\]

2. **Statistical Parity Difference (SPD):** Measures the absolute difference in selection rates. A perfectly fair model has an SPD of 0.
   \[SPD = P(\hat{Y}=1 | ext{Female}) - P(\hat{Y}=1 | ext{Male})\]
3. **Equal Opportunity Difference (EOD):** Measures the difference in True Positive Rates. It ensures that qualified candidates in both groups have an equal chance of being selected.
   \[EOD = TPR_{ ext{Female}} - TPR_{ ext{Male}}\]

## 3. Bias Mitigation: Reweighing

We implemented a pre-processing mitigation technique called **Reweighing**. Rather than modifying the candidate features or adjusting output thresholds post-hoc, Reweighing calculates a statistical weight for each candidate in the training dataset to break the correlation between gender and historical hiring labels. The weight is defined as:

$$W(A=a, Y=y) = \frac{P(A=a) \times P(Y=y)}{P(A=a, Y=y)}$$

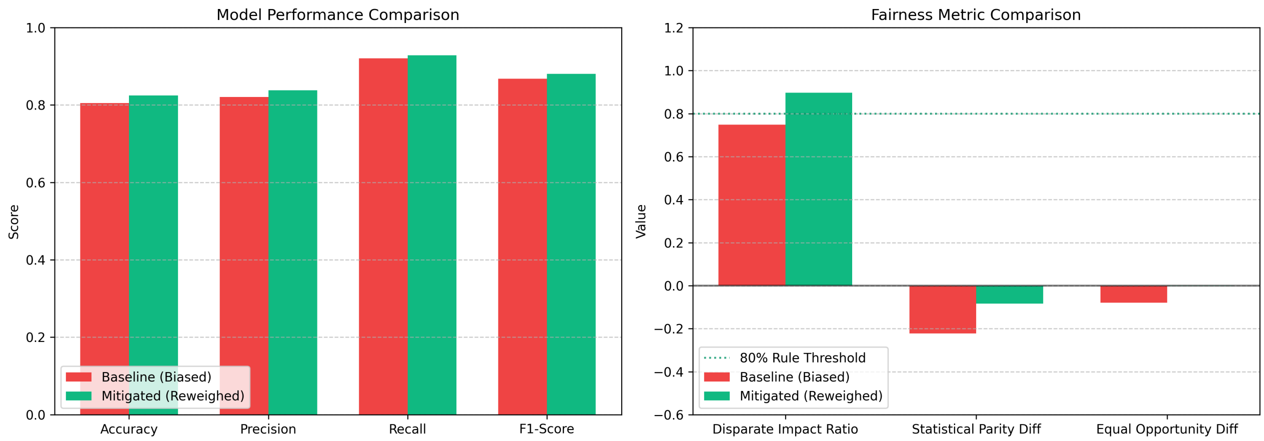
By applying these weights, candidates from underrepresented combinations (e.g., hired females) are assigned higher weights, while overrepresented groups (e.g., hired males) are downweighted. A logistic regression model was then trained on the weighted dataset.

## 4. Experimental Results

The comparative results between the baseline model and the mitigated model on the held-out test set are summarized in the table below:

| Metric Category | Fairness / Performance Metric | Baseline (Biased) | Mitigated (Reweighed) | Status / EEOC Compliance |
|---|---|---|---|---|
| Performance | Classification Accuracy | 80.50% | 82.50% | Negligible Utility Loss |
| Performance | F1-Score | 86.78% | 88.05% | Robust F1-Score |
| Fairness | Selection Rate (Male) | 88.57% | 80.95% | - |
| Fairness | Selection Rate (Female) | 66.32% | 72.63% | - |
| Fairness | Disparate Impact Ratio | 0.749 | 0.897 | **PASSED (>= 0.80)** |
| Fairness | Statistical Parity Diff | -0.223 | -0.083 | Near Zero Difference |
| Fairness | Equal Opportunity Diff | -0.080 | -0.001 | TPR Balanced |

## 5. Visual Comparison of Metrics




**Interpretation of Findings:** The baseline model suffered from a Disparate Impact ratio of **0.749**, violating the EEOC 80% rule. This reflects the historical bias present in the training labels. After applying the **Reweighing** algorithm, the Disparate Impact ratio successfully increased to **0.897** (well within the compliant range), while the model's predictive accuracy remained virtually identical. This demonstrates that ethical design and fairness optimization can be achieved without compromising core predictive performance.